UNIVERSITY OF CALIFORNIA, MERCED

CSE179: Parallel Computing

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Feb 25 and March 2

Spring 2026

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Group Communication

- Motivation: accelerate interaction patterns among groups
- Approach: collective communication
  - whole group works together *collectively* to realize a communication
  - constructed from pairwise point-to-point communications
- Implementation strategy
  - standard library of common collective operations
  - leverage target architecture for efficient implementation
- Benefits of standard library implementations
  - reduce development effort and cost for parallel codes
  - improve performance through efficient implementations
  - improve quality of scientific applications

Topics for Today

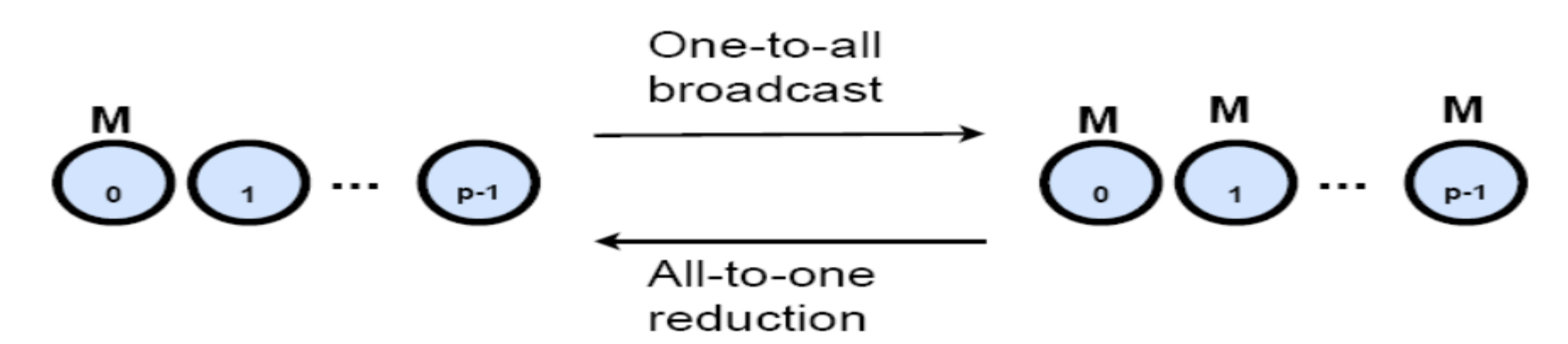
- One-to-all broadcast and all-to-one reduction
- All-to-all broadcast and reduction
- All-reduce and prefix-sum operations
- Scatter and gather
- All-to-all personalized communication
- Circular shift
- Optimizing collective patterns

Assumptions

- Network is bidirectional
- Communication is single-ported
  - node can receive only one message per step
- Communication cost model
  - message of size *m*, no congestion, time = $t_s + t_w m$
  - congestion: model by scaling t_w

One-to-All and All-to-One

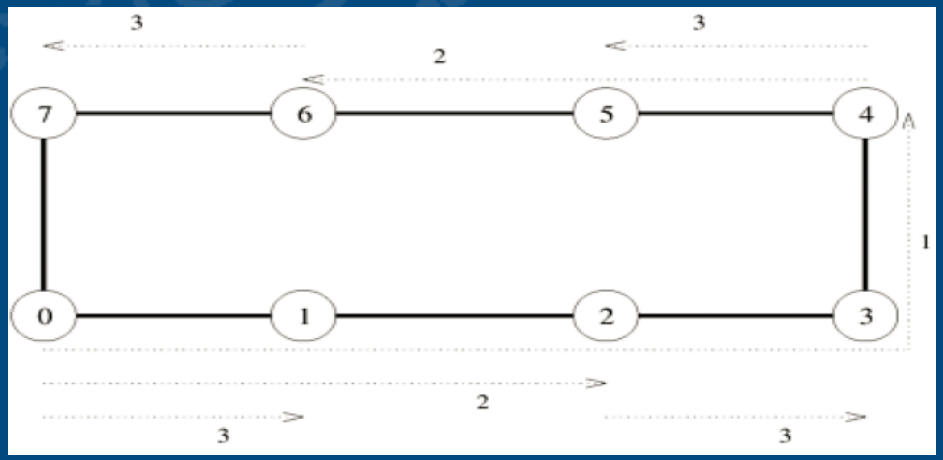
- One-to-all broadcast
  - a processor has *m* units of data that it must send to everyone
- All-to-one reduction
  - each processor has *m* units of data
  - data items must be combined using some associative operator
    - e.g. addition, min, max
  - result must be available at a target processor



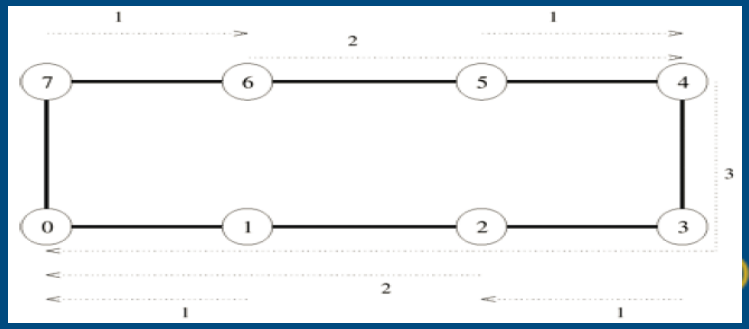
One-to-All and All-to-One on a Ring

- Broadcast
  - naïve solution
    - source sends *p - 1* messages to the other *p - 1* processors
  - use recursive doubling
    - source sends a message to a selected processor

    yields two independent problems over halves of the machine

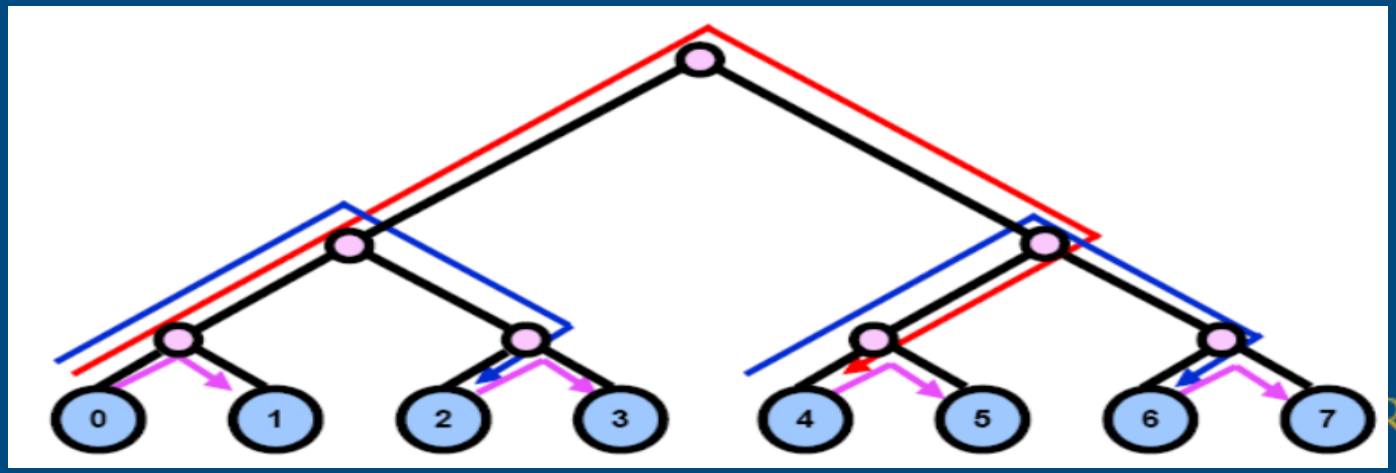


- Reduction
  - Invert the process



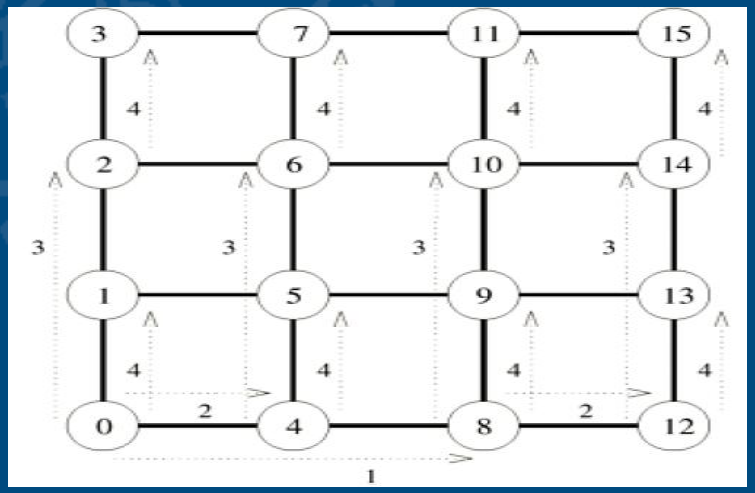
Broadcast on a Balanced Binary Tree

- Consider processors arranged in a dynamic binary tree
  - processors are (logically) at the leaves
  - internal nodes are routing nodes
- Assume leftmost processor is the root of the broadcast
- Use recursive doubling strategy: log *p* stages



Broadcast and Reduction on a 2D Mesh

- Consider a mesh of p nodes
  - view each row as a linear array of √*p* nodes
  - view each column as a linear array of √*p* nodes
- Two step broadcast and reduction operations



- Generalizes to higher dimensional meshes
  1. perform the operation along a row
  2. perform the operation along each column concurrently

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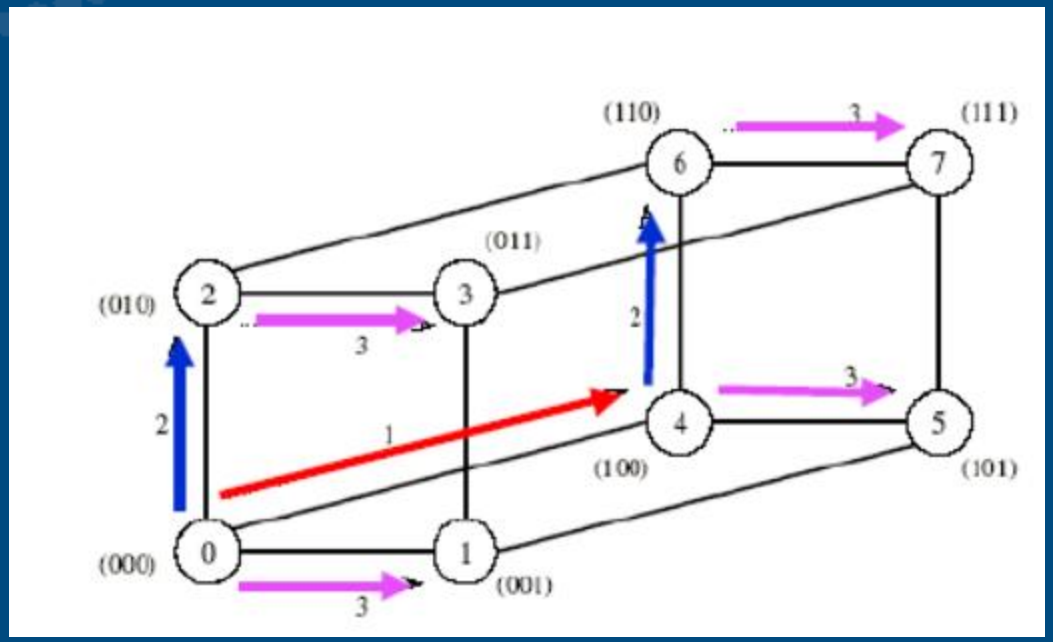
Broadcast and Reduction on a Hypercube

- Consider hypercube with 2^d nodes
  - view as *d*-dimensional mesh with two nodes in each dimension
- Apply mesh algorithm to a hypercube
  - *d* (= *log p*) steps



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Broadcast and Reduction Algorithms

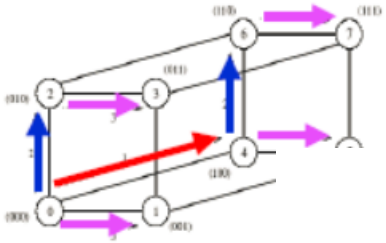
- Each of aforementioned broadcast/reduction algorithms
  - adaptation of the same algorithmic template
- Next slide: a broadcast algorithm for a hypercube of 2^d nodes
  - can be adapted to other architectures
  - in the following algorithm
    - *my_id* is the label for a node
    - *X* is the message to be broadcast

One-to-All Broadcast Algorithm

```
procedure GENERAL_ONE_TO_ALL_BC(d, my_id, source, X)
begin
    my_virtual_id := my_id XOR source;
    mask := 2^d − 1;
    for i := d − 1 downto 0 do      /* Outer loop */
        mask := mask XOR 2^i;   /* Set bit i of mask to 0 */
        if (my_virtual_id AND mask) = 0 then
            if (my_virtual_id AND 2^i) = 0 then   // even
                virtual_dest := my_virtual_id XOR 2^i;
                send X to (virtual_dest XOR source);
    /* Convert virtual_dest to the label of the physical destination */
            else                                  // odd
                virtual_source := my_virtual_id XOR 2^i;
                receive X from (virtual_source XOR source);
    /* Convert virtual_source to the label of the physical source */
            endelse;
    endfor;
end GENERAL_ONE_TO_ALL_BC
```

position relative to source

I am the master of a 2^i subcube



One-to-all broadcast of a message *X* from *source* on a hypercube

All-to-One Reduction Algorithm

```
procedure ALL_TO_ONE_REDUCE(d, my_id, m, X, sum)
begin
    for j := 0 to m − 1 do sum[j] := X[j];
    mask := 0;
    for i := 0 to d − 1 do
        /* Select nodes whose lower i bits are 0 */
        if (my_id AND mask) = 0 then
            if (my_id AND 2^i) ≠ 0 then // odd
                msg_destination := my_id XOR 2^i;
                send sum to msg_destination;
            else                        // even
                msg_source := my_id XOR 2^i;
                receive X from msg_source;
                for j := 0 to m − 1 do
                    sum[j] := sum[j] + X[j];
            endelse;
        mask := mask XOR 2^i;   /* Set bit i of mask to 1 */
    endfor;
end ALL_TO_ONE_REDUCE
```

I am the master of a 2^i subcube

All-to-One sum reduction on a *d*-dimensional hypercube

Each node contributes msg *X* containing *m* words, and node 0 is the destination

Broadcast/Reduction Cost Analysis

Hypercube

- Logp point-to-point simple message transfers
  - each message transfer time: $t_s + t_w m$

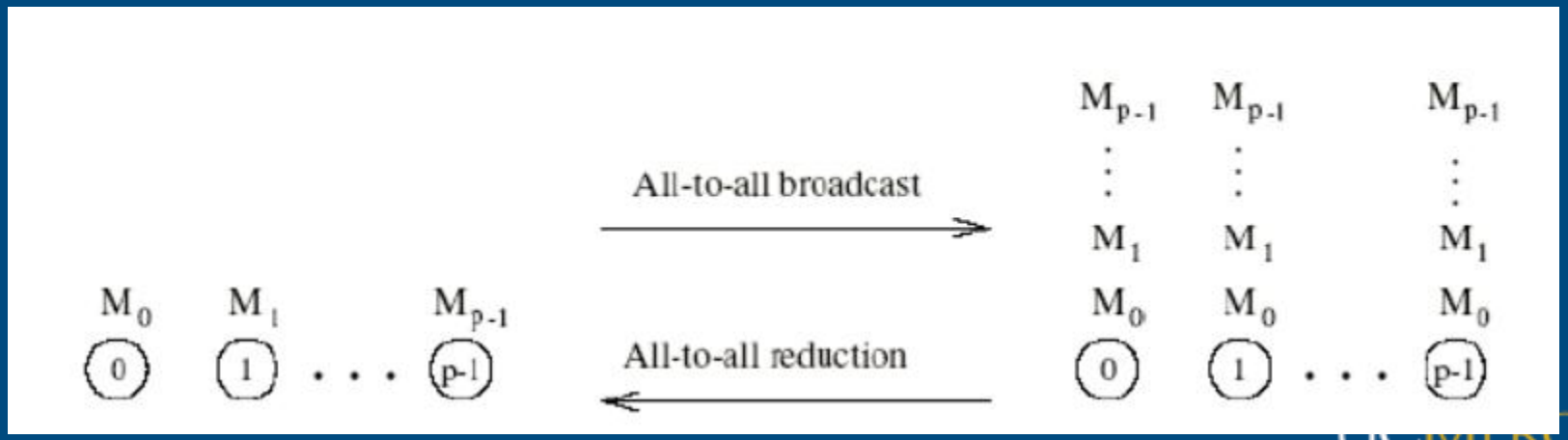
- Total time

$$T = (t_s + t_w m) \log p.$$

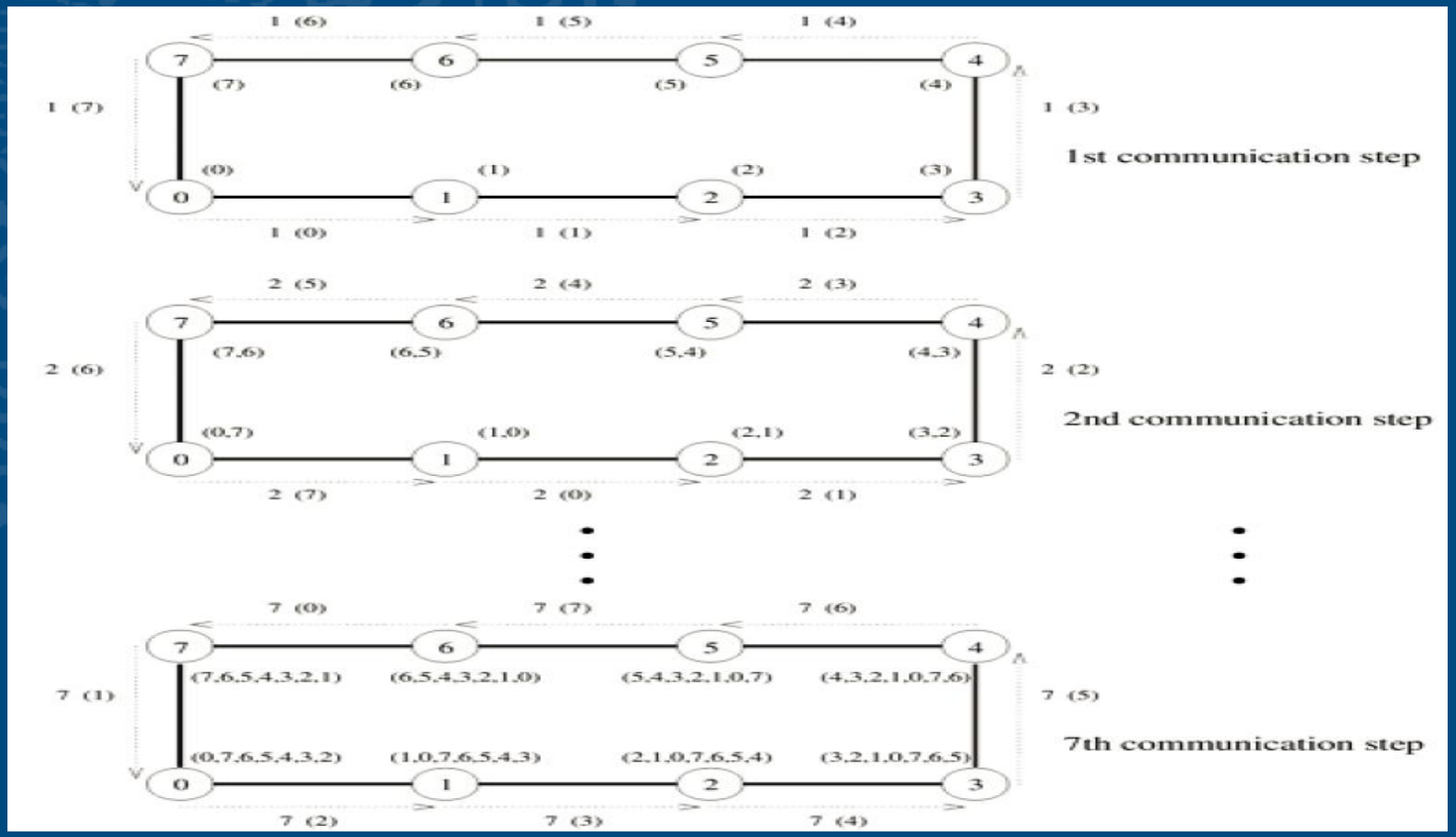
All-to-All Broadcast and Reduction

Each processor is the source as well as destination

- Broadcast
  - each process broadcasts its own *m*-word message to all others
- Reduction
  - each process gets a copy of the result



All-to-All Broadcast/Reduction on a Ring



- Communication is performed circularly in a single direction
- Each node receives all (p-1) piece of info from all other nodes in (p-1) steps

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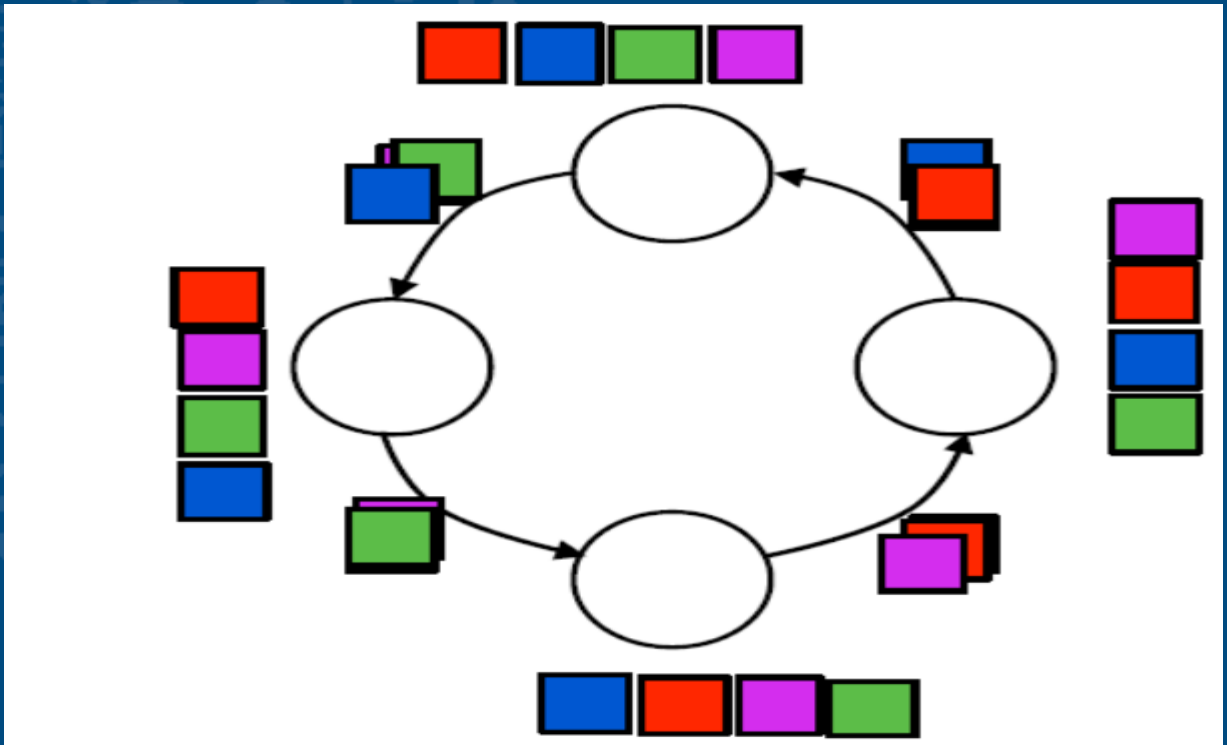
All-to-All Broadcast/Reduction on a Ring

```
procedure ALL_TO_ALL_BC_RING(my_id, my_msg, p, result)
begin
    left := (my_id − 1) mod p;
    right := (my_id + 1) mod p;
    result := my_msg;
    msg := result;
    for i := 1 to p − 1 do
        send msg to right;
        receive msg from left;
        result := result ∪ msg;
    endfor;
end ALL_TO_ALL_BC_RING
```

message size stays constant

All-to-All Broadcast on a Ring



For an all-to-all reduction,

- combine (rather than append) each incoming message into your local result
- forward to your successor what you have received

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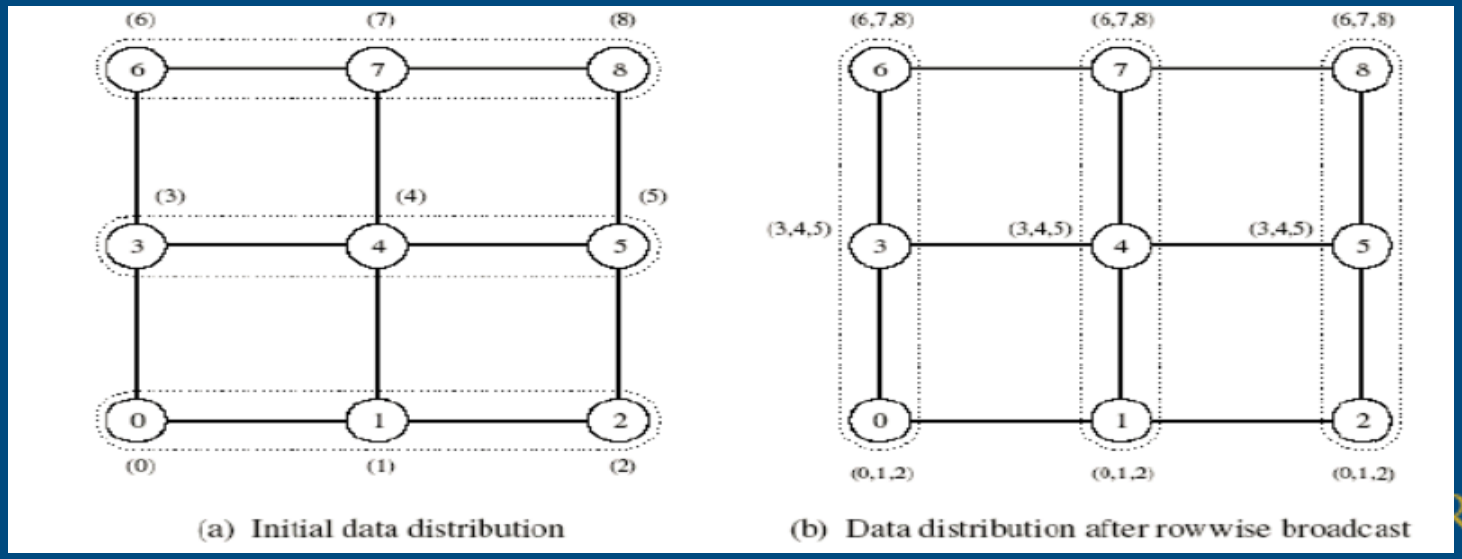
All-to-all Broadcast on a Mesh

Two phases

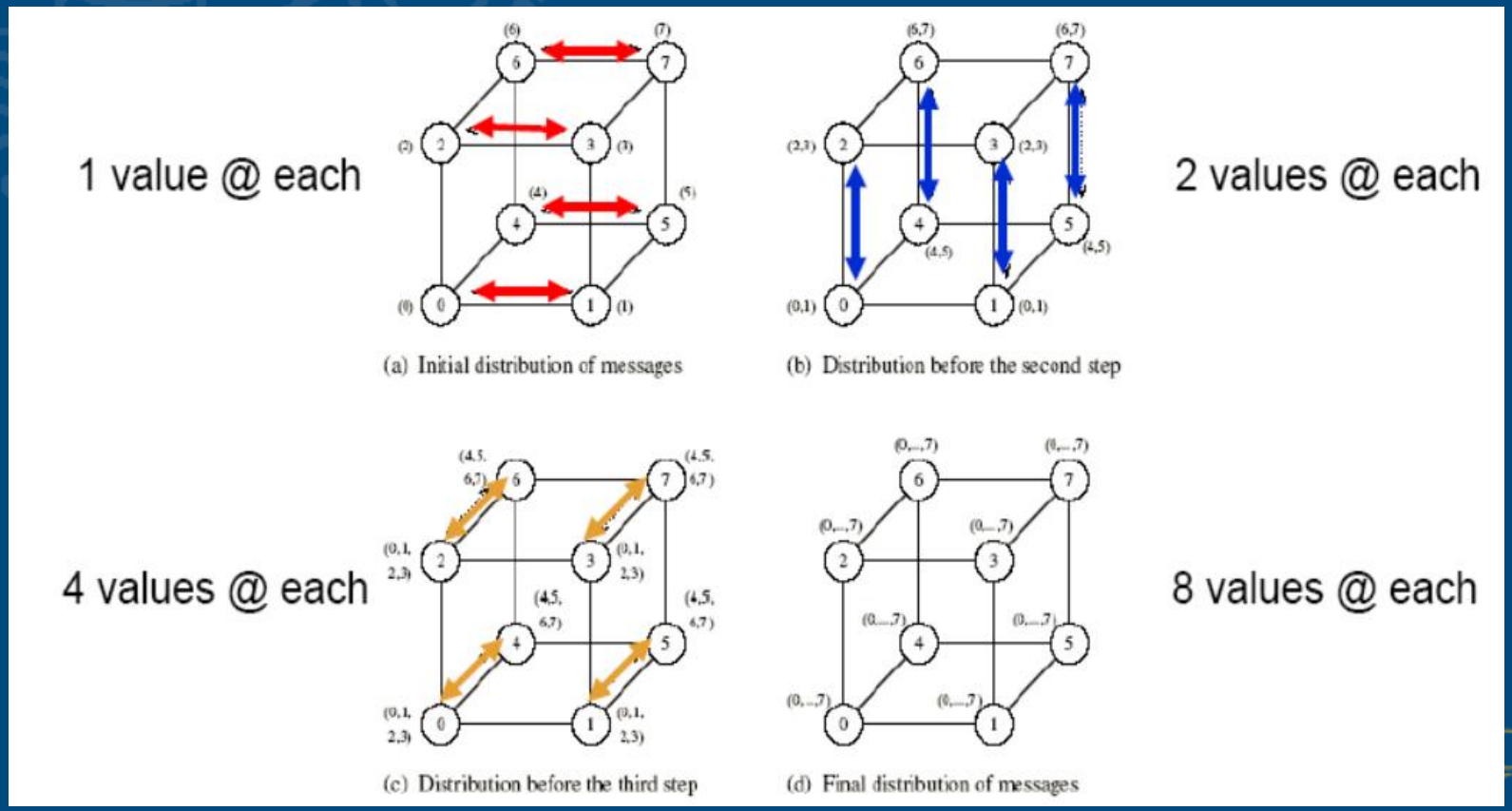
- Each row performs all-to-all broadcast as for linear array/ring
  - each node collects √*p -1* messages for nodes in its own rows
  - consolidates into a single message of size *m*√p.
- Perform column-wise all-to-all broadcast of merged messages



(a) Initial data distribution (b) Data distribution after rowwise broadcast

All-to-all Broadcast on a Hypercube

- Generalization of the mesh algorithm to *log p* dimensions
- Message size doubles in each of *log p* steps



All-to-all Broadcast on a Hypercube

```
procedure ALL_TO_ALL_BC_HCUBE(my_id, my_msg, d, result)
begin
    result := my_msg;
    for i := 0 to d − 1 do
        partner := my_id XOR 2^i;
        send result to partner;
        receive msg from partner;
        result := result ∪ msg;
    endfor;
end ALL_TO_ALL_BC_HCUBE
```

All-to-all Reduction

- Similar to all-to-all broadcast, except for the merge
- Algorithm sketch

```
my_result = local_value

for each round
   send my_result to partner
   receive msg
   my_result = my_result ⊕ msg

post condition: each my_result now contains global result
```

All-to-all Reduction

- Similar to all-to-all broadcast, except for the merge
- Algorithm sketch

```
procedure ALL_TO_ALL_RED_HCUBE(my_id, msg, d, result)
begin
    recloc := 0;
    for i := d - 1 to 0 do
        partner := my_id XOR 2^i;
        j := my_id AND 2^i;
        k := (my_id XOR 2^i) AND 2^i;
        senloc := recloc + k;
        recloc := recloc + j;
        send msg[senloc .. senloc + 2^i - 1] to partner;
        receive temp[0 .. 2^i - 1] from partner;
        for j := 0 to 2^i - 1 do
            msg[recloc + j] := msg[recloc + j] + temp[j];
        endfor;
    endfor;
    result := msg[my_id];
end ALL_TO_ALL_RED_HCUBE
```

Cost Analysis for All-to-All Broadcast

- Ring

  $(t_s + t_w m)(p-1)$

- Mesh
  - phase 1: $(t_s + t_w m)(\sqrt{p} - 1)$
  - Phase 2: $(t_s + t_w m\sqrt{p})(\sqrt{p} - 1)$
  - Total: $2t_s(\sqrt{p} - 1) + t_w m(p-1)$

- Hypercube

$$T = \sum_{i=1}^{\log p} (t_s + 2^{i-1} t_w m)$$

$$= t_s \log p + t_w m(p-1).$$

Above algorithms are asymptotically optimal in msg size

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Prefix Sum

- Pre-condition
  - given p numbers $n_0, n_1, \ldots, n_{p-1}$ (one on each node)
    - The node labeled k contains n_k
- Problem statement
  - compute the sums $s_k = \sum_{i=0}^{k} n_i$ for all k between 0 and $p-1$
- Post-condition
  - The node labeled k contains S_k

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Prefix Sum

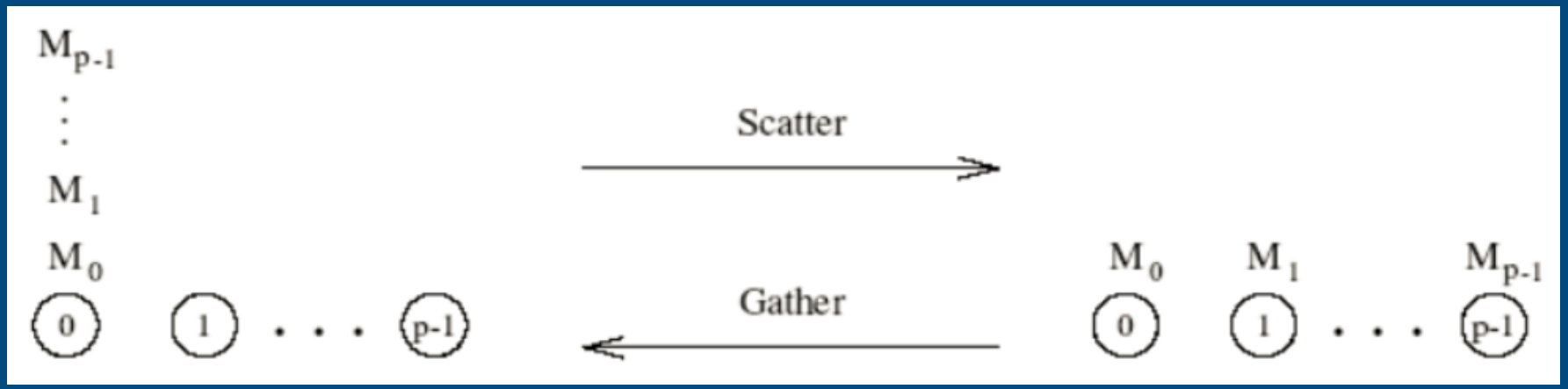
- Can implement using all-to-all reduction kernel
- Constraint
  - prefix sums on node k: values from k-node subset with labels $\leq k$
- Strategy
  - implemented using an additional result buffer
  - add incoming value to the result buffer on node k
    - only if the msg from a node $\leq k$

Prefix Sum on a Hypercube

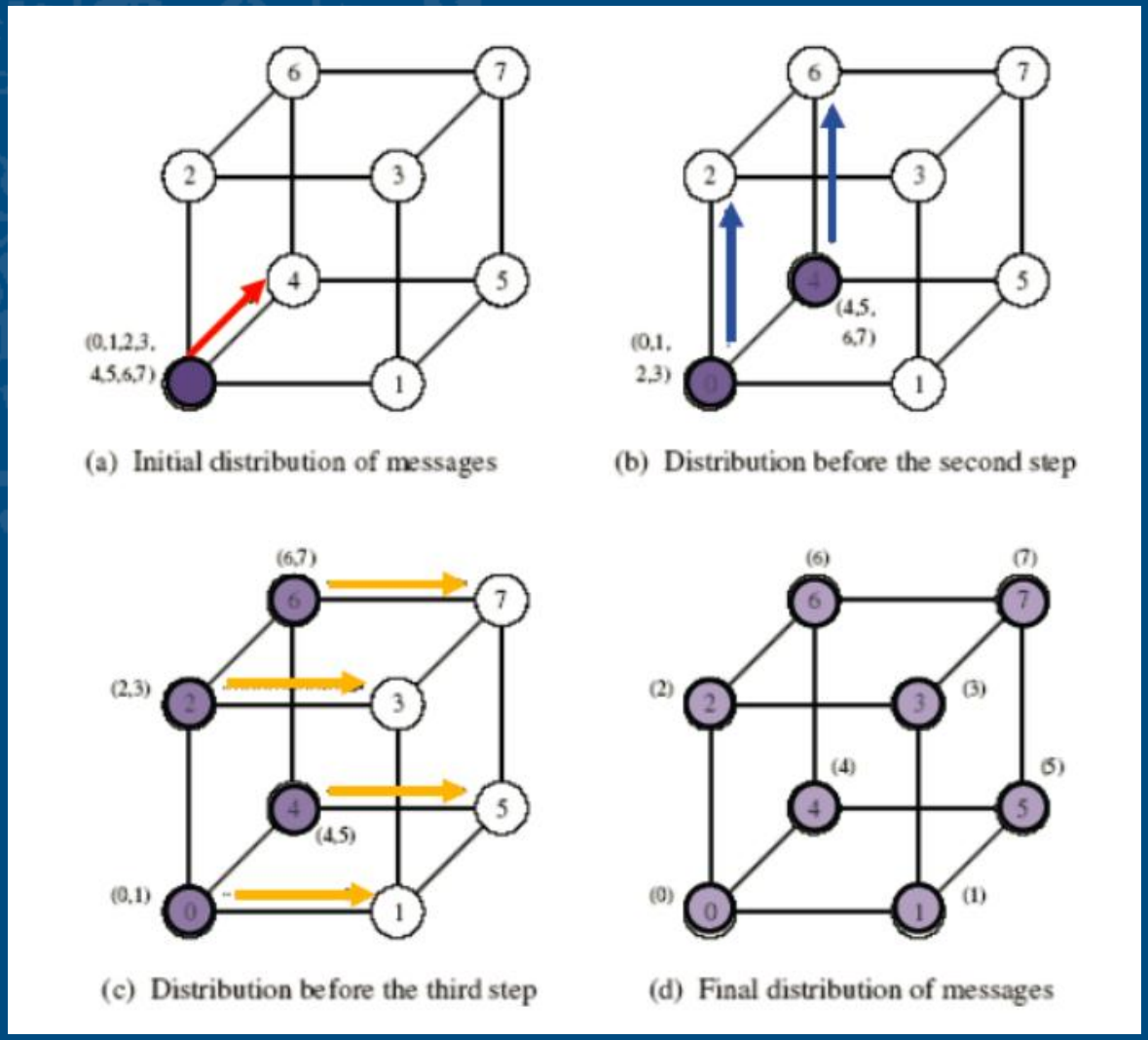
```
procedure PREFIX_SUMS_HCUBE(my_id, my_number, d, result)
begin
    result := my_number;
    msg := result;
    for i := 0 to d − 1 do
        partner := my_id XOR 2^i;
        send msg to partner;
        receive number from partner;
        msg := msg + number;
        if (partner < my_id) then result := result + number;
    endfor;
end PREFIX_SUMS_HCUBE
```

Scatter and Gather

- Scatter
  - a node sends a unique message of size *m* to every other node
  - AKA one-to-all personalized communication
  - algorithmic structure is similar to broadcast
    - scatter: message size get smaller at each step
    - broadcast: message size stay constant
- Gather
  - single node collects a unique message from each node
  - inverse of the scatter operation; can be executed as such



Scatter on a Hypercube



(a) Initial distribution of messages

(b) Distribution before the second step

(c) Distribution before the third step

(d) Final distribution of messages

Cost of Scatter and Gather

- Log *p* steps
  - in each step
    - machine size doubles
    - message size halves
- Time

$$T = t_s \log p + t_w m(p-1)$$

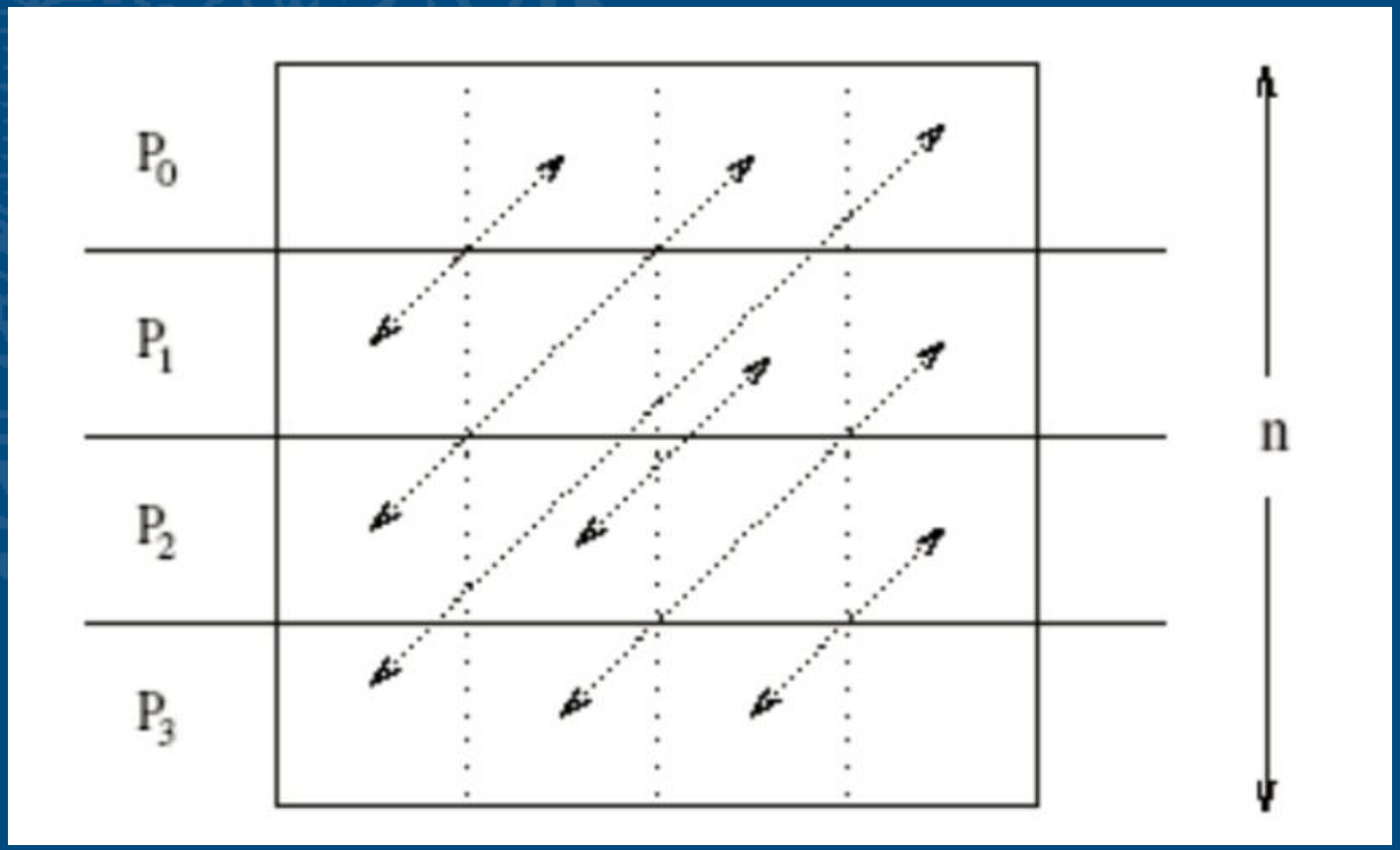
- Note: time is asymptotically optimal in message size

All-to-All Personalized Communication

Total exchange

- Each node: distinct message of size *m* for every other node

All-to-All Personalized Communication



- All-to-all personalized communication in transposing a matrix using four processes

All-to-All Personalized Communication

- Every node has p pieces of data, each of size m
- Algorithm sketch for a ring

```
for k = 1 to p - 1
   send message of size m(p - k) to neighbor
   select piece of size m out of message for self
```

- Cost analysis

$$
\begin{aligned}
T &= \sum_{i=1}^{p-1} (t_s + t_w m(p-i)) \\
&= t_s(p-1) + t_w m \sum_{i=1}^{p-1} i \\
&= (t_s + t_w mp/2)(p-1)
\end{aligned}
$$

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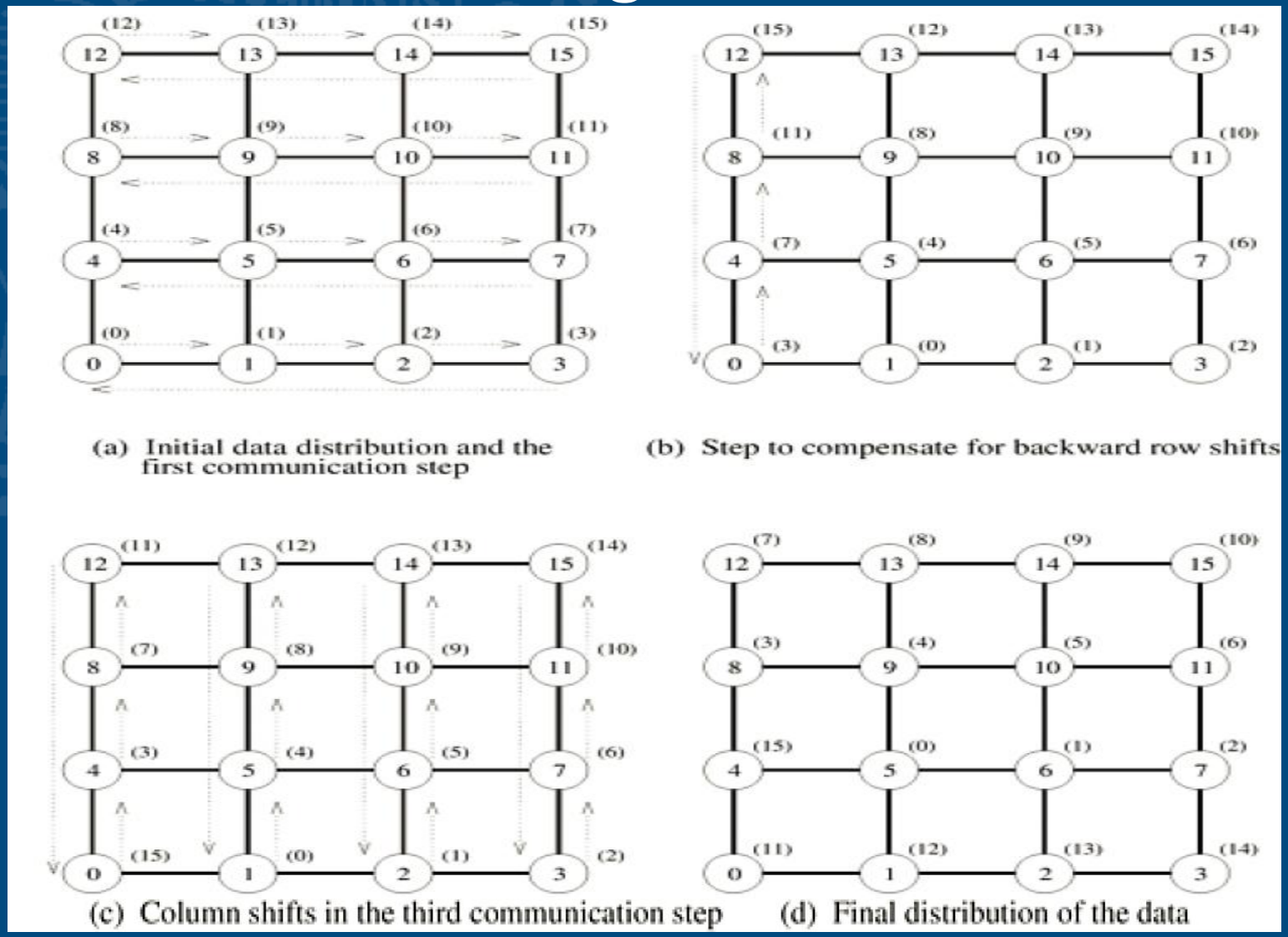
Circular Shift

- Permutation on p nodes
  - node *i* sends a data packet to node *(i + q)* mod *p, (0 < q < p)*
- Ring
  - requires min(*q, p – q*) neighbor communications
- Torus
  - collectively shift along each direction in phases
  - time upper bound

$$T = (t_s + t_w m)(\sqrt{p} + 1).$$

Circular Shift on a Ring Embedded in a Torus



- Circular 5-shift on a 4x4 torus

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Optimizing Collective Patterns

- Consider broadcast of message *M* of size *m*, where *m* is large
- Cost of straightforward strategy

$$T = (t_s + t_w m)\log p$$

- Optimized strategy
  - split *M* into *p* parts *M0, M1, … Mp-1* of size *m/p* each
    - want to place *M0* ∪ *M1* ∪ … ∪ *Mp-1* on all nodes
  - scatter *Mi* to node *i*
  - have nodes collectively perform all-to-all broadcast
    - each node *k* broadcasts its *Mk*
- Cost analysis
  - scatter time = $t_s \log p + t_w(m/p)(p-1)$ *(slide 29)*
  - all-to-all broadcast time = $t_s \log p + t_w(m/p)(p-1)$ *(slide 23)*
  - total time = $2(t_s \log p + t_w(m/p)(p-1)) \approx 2(t_s \log p + t_w m)$

(faster than slide 13)

$$T = (t_s + t_w m)\log p.$$

Reading Materials

- Chapter 4
  - 4.1-4.8

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